

Data Structure and Algorithms

What is Data Structure and Algorithm:-

Data structure:-

It is a data organization, management, and storage format that enables efficient access and modification.

⇒ More precisely, a data structure is a collection of data values, the relationships among them, and the functions or operations that can be applied to the data.

Algorithm:-

It is a combination of sequence of finite steps to solve a particular problem.

for example:- Multiplication of two numbers

MULT

1. Take two numbers (a, b)

2. Multiply two numbers a and b and store the value of result

3. return c

properties of Algorithms:-

- ⇒ It should terminate after finite time.
- ⇒ It should produce at least one output.
- ⇒ It is independent of any sort of programming language.
- ⇒ It should be Deterministic

Deterministic - for same input same output will come always

non Deterministic - for the same input different output will come

and this is not all preferable whenever we write any sort of algorithms.

Data structures and Algorithms topics :-

1. Arrays (Introduction, Basic operations, Searching and sorting algorithms, Standard problems)
2. Strings
3. Hashing
4. Stack
5. Queue
6. pointers
7. Linked list
8. Tree, Segment Tree, Trie
9. Heap
10. Graph Traversal

Algorithms :-

1. Time and space complexity
2. Recurrence Relation solving
3. Recursion
4. Divide and conquer
 - ⇒ finding of Max and Min
 - ⇒ finding of power of an element
 - ⇒ Searching → Linear search, Binary search
 - ⇒ Sorting → Mergesort, Quicksort, selection sort, Insertion sort, Bubble sort, Radix sort, Count sort, Bucket sort,
 - ⇒ Heapsort → Minheap, max heap
5. Greedy Techniques
 - ⇒ Knapsack problem
 - ⇒ Job sequencing with deadline
 - ⇒ Huffman coding
 - ⇒ Optimal merge pattern
 - ⇒ Minimum cost spanning Tree : Kruskal, Prim's
 - ⇒ Single source shortest path : Dijkstra's Algorithm, Bellman ford

6. Dynamic programming

- ⇒ fibonacci series
- ⇒ longest common subsequence
- ⇒ 0/1 knapsack
- ⇒ sum of subset
- ⇒ All shortest path
- ⇒ Matrix chain Multiplication

loops in java :-

⇒ loops are used to execute a set of Instructions repeatedly

1) for-loop :-

It is used to Iterate the part of program several times.
if number of Iterations are fixed.

2) while loop :- used to iterate a part of the program several times. If the number of iterations are not fixed, It is recommended to go for "while loop".

3) do-while loop :- It is used to Iterate a part of the program several times. If the number of iterations are not fixed and it will execute the loop atleast once.